# Shopping List

Submersible Pump Power Abnormality Monitoring & Protection System

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This list is tailored to your existing Hyundai 3-phase DOL starter panel, the parts that can be salvaged from the 2008 office PCs, and a low-budget approach. Items marked “Essential” must be obtained. Checkboxes are provided for tracking.

## 1. Essential Items (Must Obtain)

| ☐ | Qty | Item | Recommended Type / Notes | Purpose |
|---|---|---|---|---|
| ☐ | 1 | **ESP32 development board** | ESP32 DevKit V1 or ESP32-WROOM-32 (USB-C preferred) | Main controller (dual-core, ADC, Wi-Fi) |
| ☐ | 3 | **Voltage sensor** | ZMPT101B module (isolated) | Measure each phase voltage safely |
| ☐ | 3 | **Current sensor** | SCT-013-030 (30 A, 1 V output, clamp-on) | Non-invasive current measurement – safest for live work |
| ☐ | 1 | **Relay module** | 1- or 2-channel 5 V opto-isolated relay | Switches the contactor coil to trip the pump |
| ☐ | 1 | **Display** | 0.96″ or 1.3″ I2C OLED (SSD1306) | Show voltages, currents, status & trip reason |
| ☐ | 1 | **Reset push-button** | Momentary normally-open (panel mount preferred) | Manual reset after a protection trip (required) |
| ☐ | 3–5 | **Status LEDs + resistors** | Red / Yellow / Green + 220–330 Ω resistors | Power, Running, Fault / Warning indication |
| ☐ | 1 set | **Prototyping hardware** | Breadboard or perfboard + screw terminal blocks | Clean, reliable sensor and control connections |

## 2. Power Supply (Choose One)

| ☐ | Option | Item | Notes |
|---|---|---|---|
| ☐ | **Preferred (free)** | Salvaged ATX PSU from old company PC | Use the +5 V rail after testing. Maintain galvanic isolation from the high-voltage motor side. |
| ☐ | **Alternative** | Small isolated 5 V supply (Hi-Link 5V/5W or equivalent) | Cleaner and more compact if you prefer not to reuse the old PSU. |

## 3. Strongly Recommended (Improves Quality & Professionalism)

| ☐ | Qty | Item | Notes |
|---|---|---|---|
| ☐ | 1 | **ADS1115 16-bit ADC module** | Much better accuracy and resolution than the ESP32 internal ADC. Highly recommended for reliable protection thresholds. |
| ☐ | 1 | **Small DIN-rail enclosure or metal box** | House the ESP32 + sensors neatly next to your existing panel (or reuse the salvaged PC case). |
| ☐ | — | **Extra terminal blocks, ferrules, heat-shrink, cable ties** | Professional, reliable wiring – worth the small extra cost. |

## 4. Already Available (Do Not Buy)

- Your existing Hyundai contactor + thermal overload panel (actuation layer)
- Old PC metal case / chassis (enclosure)
- Fans, screws, standoffs, and good wires from the salvaged PCs
- Possible spare terminal blocks or CTs from company stock

## 5. Important Notes Before Buying

• **Contactor coil voltage** — Check the coil voltage printed on your Hyundai contactor (common values: 24 V DC, 110 V AC, 220 V AC…). The relay module must be rated to switch that voltage safely. Confirm this before purchasing the relay.

• **SCT-013-030 preferred** — Non-invasive clamp-on design is much safer than series Hall sensors (ACS712) when working on a live industrial pump.

• **Isolation** — Always keep the high-voltage motor side galvanically isolated from the ESP32 and sensor electronics.

• **Buy a few extras** of the cheap items (LEDs, resistors, terminal blocks). They are always useful during development and final assembly.

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*Once the essential items are obtained, the next step is Stage 2 (Hardware Design) and integration with your existing contactor panel.*